# A hybridized DG method with LMARS interface flux for the linearized Euler equations with gravity, with relaxed trace conditions and time integration without elimination

Documentation of `hdg_relax_euler_gravity.py` / `HDGRelaxEuler.jl`

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**Abstract**

We describe a one-dimensional hybridized discontinuous Galerkin (HDG) discretization of the compressible Euler equations with gravity, linearized about a hydrostatic background at rest. The single-valued face traces are fixed by additional algebraic conditions (conservativity at interior faces, reflection at walls). With the stabilization $\boldsymbol{T} = |\boldsymbol{A}|$ the resulting face flux is exactly the linearized LMARS flux. Instead of eliminating the element unknowns in favour of the traces (static condensation), the additional conditions are *relaxed* into evolution equations $\varepsilon\,\partial_t \hat{\boldsymbol{q}} = \boldsymbol{H}\boldsymbol{q} - \hat{\boldsymbol{q}}$, and the coupled element/trace system is integrated in time directly, either monolithically with a stiffly accurate SDIRK method or with an IMEX scheme whose implicit stage is local to each face. We prove well-balancing and discuss conservation and the $\mathcal{O}(\varepsilon c/h)$ relaxation error, and report verification results.

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# 1 Model

Let $z$ be the vertical coordinate. The background state is at rest and in hydrostatic balance,

$$u_0 = 0, \qquad \partial_z p_0 = -\rho_0 g, \qquad c_0^2(z) = \gamma \frac{p_0}{\rho_0} = \gamma R T_0(z). \tag{1}$$

The code uses a constant lapse rate $T_0(z) = T_{00} - \Gamma z$ (isothermal for $\Gamma = 0$). Linearizing the Euler equations around this state for perturbations $(\rho', u', p')$ gives

$$\partial_t \rho' + \partial_z(\rho_0 u') = 0, \tag{2}$$
$$\rho_0 \partial_t u' + \partial_z p' = -g\rho', \tag{3}$$
$$\partial_t p' + u'\,\partial_z p_0 + \gamma p_0\,\partial_z u' = 0. \tag{4}$$

With the momentum perturbation $m = \rho_0 u'$ and the identity $\partial_z(c_0^2 m) = \gamma \partial_z(p_0 u') = \gamma p_0 \partial_z u' + \gamma u' \partial_z p_0$ together with $\partial_z p_0 = -\rho_0 g$, the pressure equation becomes $\partial_t p' + \partial_z(c_0^2 m) = -(\gamma - 1) g\, m$. Hence, with $\boldsymbol{q} = (\rho', m, p')^{\mathsf{T}}$,

$$\boxed{\partial_t \boldsymbol{q} + \partial_z\big(\boldsymbol{A}(z)\boldsymbol{q}\big) = \boldsymbol{S}\boldsymbol{q}, \qquad \boldsymbol{A} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & c_0^2 & 0 \end{pmatrix}, \quad \boldsymbol{S} = \begin{pmatrix} 0 & 0 & 0 \\ -g & 0 & 0 \\ 0 & -(\gamma-1)g & 0 \end{pmatrix}.} \tag{5}$$

The background density $\rho_0$ does not appear; only $c_0^2(z)$ does. The eigenvalues of $\boldsymbol{A}$ are $0, \pm c_0$. The zero-speed characteristic is pure $\rho'$ (right eigenvector $(1, 0, 0)^{\mathsf{T}}$), and the acoustic part acts on $(m, p')$.

**Hydrostatic perturbations.** Every state with $m = 0$ and $\partial_z p' = -g\rho'$ is steady. A scheme is called *well-balanced* if it preserves the discrete analogue exactly.

# 2 HDG discretization

## 2.1 Mesh, spaces, local problem

The interval $[z_{\min}, z_{\max}]$ is split into $N_e$ elements $K_e = [z_{e-1/2}, z_{e+1/2}]$ of size $h$. On each element $\boldsymbol{q}_h \in P_p(K_e)^3$, expanded in Legendre polynomials (modal basis, diagonal mass matrix). On each face $f$ there is a single-valued trace $\hat{\boldsymbol{q}}_f \in \mathbb{R}^3$. The local problem reads: for all $\boldsymbol{v} \in P_p(K)^3$,

$$(\boldsymbol{v}, \partial_t \boldsymbol{q}_h)_K = (\partial_z \boldsymbol{v}, \boldsymbol{A}\boldsymbol{q}_h)_K + (\boldsymbol{v}, \boldsymbol{S}\boldsymbol{q}_h)_K - \sum_{f \in \partial K} \boldsymbol{v}(z_f)^{\mathsf{T}}\, \widehat{\boldsymbol{F}}_{K,f}, \tag{6}$$

with the one-sided HDG flux ($n = \pm 1$ the outward normal)

$$\widehat{\boldsymbol{F}}_{K,f} = \boldsymbol{A}(z_f)\, \hat{\boldsymbol{q}}_f\, n + \boldsymbol{T}_{K,f}\, \big(\boldsymbol{q}_h|_K(z_f) - \hat{\boldsymbol{q}}_f\big). \tag{7}$$

The volume integrals are evaluated with $(p+3)$-point Gauss quadrature, which integrates $(\partial_z \boldsymbol{v}, c_0^2 m)$ exactly for linear $c_0^2$.

### 2.2 Additional (trace) conditions

The traces are determined by algebraic conditions on each face.

- **Interior face** (left element $L$, right element $R$): conservativity,

$$\widehat{\boldsymbol{F}}_{L,f} + \widehat{\boldsymbol{F}}_{R,f} = \boldsymbol{T}_{L,f}(\boldsymbol{q}_L - \hat{\boldsymbol{q}}_f) + \boldsymbol{T}_{R,f}(\boldsymbol{q}_R - \hat{\boldsymbol{q}}_f) = \boldsymbol{0}. \tag{8}$$

- **Wall face** (reflecting): mirror average with the ghost state $(\rho', -m, p')$,

$$\hat{\boldsymbol{q}}_f = \boldsymbol{P}\boldsymbol{q}_{\text{int}}, \qquad \boldsymbol{P} = \operatorname{diag}(1, 0, 1) \quad \Rightarrow \quad \hat{m}_f = 0. \tag{9}$$

Both are written as $\hat{\boldsymbol{q}} = \boldsymbol{H}\boldsymbol{q}$, with $\boldsymbol{H}$ a sparse, face-local averaging operator.

### 2.3 LMARS stabilization

For the linear system the eigenvalues of $\boldsymbol{A}$ are $\{0, \pm c\}$, so $|\boldsymbol{A}| = \boldsymbol{A}^2/c$, i.e.

$$\boldsymbol{T}_{\text{LMARS}}(z_f) = |\boldsymbol{A}(z_f)| = \begin{pmatrix} 0 & 0 & 1/c_f \\ 0 & c_f & 0 \\ 0 & 0 & c_f \end{pmatrix}, \qquad c_f = c_0(z_f). \tag{10}$$

Because the background is continuous, $\boldsymbol{T}_{L,f} = \boldsymbol{T}_{R,f}$, and (8) gives $|\boldsymbol{A}|(\boldsymbol{q}_L + \boldsymbol{q}_R - 2\hat{\boldsymbol{q}}) = 0$. The second and third rows give $\hat{m} = \{\{m\}\}$ and $\hat{p} = \{\{p'\}\}$. The first column of $\boldsymbol{A}$ and of $|\boldsymbol{A}|$ vanishes, so $\hat{\rho}$ never enters a flux and we set $\hat{\rho} = \{\{\rho'\}\}$. Inserting $\hat{\boldsymbol{q}} = \{\{\boldsymbol{q}\}\}$ into (7) (left side, $n = +1$, $[\![\cdot]\!] = (\cdot)_R - (\cdot)_L$):

$$\widehat{\boldsymbol{F}} = \begin{pmatrix} m^* \\ p^* \\ c_f^2 m^* \end{pmatrix}, \qquad m^* = \{\{m\}\} - \frac{[\![p']\!]}{2c_f}, \qquad p^* = \{\{p'\}\} - \frac{c_f}{2}[\![m]\!]. \tag{11}$$

This is exactly the LMARS flux of Chen et al. [1] linearized about $u_0 = 0$, with interface velocity $u^* = m^*/\rho_0$ and interface pressure $p^*$. It is also the exact Godunov flux of the acoustic subsystem. The flux identity is checked numerically in the code (to round-off).

**Alternative (LLF).** The option `flux="llf"` uses $\boldsymbol{T}_K = c_K \operatorname{diag}(\tau_\rho, 1, 1)$ with an element value $c_K$, which gives a $c_K$-weighted mean as the trace. With $\tau_\rho = 0$ it is well-balanced, but $\rho'$ then converges only at order $p$ (Table 1). With $\tau_\rho > 0$ it penalizes the discontinuous $\rho'$ of a hydrostatic state, which destroys exact well-balancing. LMARS avoids both problems: it penalizes only pressure and momentum jumps, and couples the mass flux to the pressure jump.

### 2.4 Matrix form

Collecting all element coefficients in $\boldsymbol{q} \in \mathbb{R}^{n_q}$ ($n_q = 3N_e(p+1)$) and all traces in $\hat{\boldsymbol{q}} \in \mathbb{R}^{n_t}$ ($n_t = 3N_f$), (6)–(9) read

$$\boldsymbol{M}\dot{\boldsymbol{q}} = \boldsymbol{K}\boldsymbol{q} + \boldsymbol{G}\hat{\boldsymbol{q}}, \qquad \boldsymbol{0} = \boldsymbol{H}\boldsymbol{q} - \hat{\boldsymbol{q}}, \tag{12}$$

where $\boldsymbol{M}$ is block diagonal (here diagonal), $\boldsymbol{K}$ contains the volume, source and element parts of the penalty, $\boldsymbol{G}$ couples the traces into the element equations, and $\boldsymbol{H}$ is the trace condition. The standard HDG implementation would eliminate $\boldsymbol{q}$ (static condensation) and solve a global system for $\hat{\boldsymbol{q}}$ only. That is not done here.

# 3 Relaxation of the additional conditions

The algebraic condition is replaced by a relaxation equation with relaxation time $\varepsilon \geq 0$:

$$\boxed{\begin{pmatrix} \boldsymbol{M} & \boldsymbol{0} \\ \boldsymbol{0} & \varepsilon \boldsymbol{I} \end{pmatrix} \frac{d}{dt} \begin{pmatrix} \boldsymbol{q} \\ \hat{\boldsymbol{q}} \end{pmatrix} = \underbrace{\begin{pmatrix} \boldsymbol{K} & \boldsymbol{G} \\ \boldsymbol{H} & -\boldsymbol{I} \end{pmatrix}}_{\boldsymbol{L}} \begin{pmatrix} \boldsymbol{q} \\ \hat{\boldsymbol{q}} \end{pmatrix}.} \tag{13}$$

For $\varepsilon = 0$ this is the index-1 DAE (12), i.e. the HDG/LMARS scheme. For $\varepsilon > 0$ the traces lag behind the element states. The system is integrated as a whole, and no Schur complement is formed.

**Relaxation error.** Expanding the slow manifold, $\hat{\boldsymbol{q}} = \boldsymbol{H}\boldsymbol{q} - \varepsilon\, \boldsymbol{H}\dot{\boldsymbol{q}} + \mathcal{O}(\varepsilon^2)$. The trace defect enters the element equations through $\boldsymbol{G}$, whose entries scale like $|\boldsymbol{T}| \sim c$, while $\boldsymbol{M} \sim h$. The perturbation of the semi-discrete operator is therefore

$$\boldsymbol{M}^{-1}\boldsymbol{G}\,(\hat{\boldsymbol{q}} - \boldsymbol{H}\boldsymbol{q}) = \mathcal{O}\Big(\frac{c\,\varepsilon}{h}\Big)\,\dot{\boldsymbol{q}}, \tag{14}$$

i.e. a relative consistency error of order $\varepsilon c/h$. This scaling is confirmed by Table 2. To retain the HDG accuracy one needs $\varepsilon c/h \ll$ (spatial error), e.g. $\varepsilon \lesssim 10^{-7} h/c$ for $p = 3$ on the test mesh. The implicit or local treatment of the relaxation term makes such small $\varepsilon$ affordable.

# 4 Time integration without elimination

## 4.1 Monolithic SDIRK (`method="dirk"`)

We use the five-stage, fourth-order, L-stable and stiffly accurate SDIRK method of Hairer and Wanner [4] with $a_{ii} = 1/4$:

| | | | | | |
|---|---|---|---|---|---|
| 1/4 | 1/4 | | | | |
| 3/4 | 1/2 | 1/4 | | | |
| 11/20 | 17/50 | −1/25 | 1/4 | | |
| 1/2 | 371/1360 | −137/2720 | 15/544 | 1/4 | |
| 1 | 25/24 | −49/48 | 125/16 | −85/12 | 1/4 |
| | 25/24 | −49/48 | 125/16 | −85/12 | 1/4 |

With $\mathcal{M} = \mathrm{diag}(\boldsymbol{M}, \varepsilon\boldsymbol{I})$, every stage solves

$$(\mathcal{M} - \Delta t\, a_{ii}\boldsymbol{L})\,\boldsymbol{Y}_i = \mathcal{M}\boldsymbol{U}^n + \Delta t \sum_{j<i} a_{ij}\boldsymbol{L}\boldsymbol{Y}_j, \qquad \boldsymbol{U}^{n+1} = \boldsymbol{Y}_5. \tag{15}$$

The matrix $\mathcal{M} - \Delta t\, a_{ii}\boldsymbol{L}$ of the *full* coupled system is factorized once (sparse LU). It is regular for $\varepsilon = 0$ as well, because its trace block is $\Delta t\, a_{ii}\boldsymbol{I}$. Stiff accuracy makes the scheme valid for the DAE limit.

## 4.2 IMEX ARS(2,2,2) (`method="imex"`)

The element part is treated explicitly and the relaxation implicitly, using ARS(2,2,2) of Ascher, Ruuth and Spiteri [3] with $\gamma = 1 - 1/\sqrt{2}$ and $\delta = 1 - 1/(2\gamma)$:

explicit:

| | | | |
|---|---|---|---|
| 0 | 0 | | |
| $\gamma$ | $\gamma$ | 0 | |
| 1 | $\delta$ | $1-\delta$ | 0 |
| | $\delta$ | $1-\delta$ | 0 |

implicit:

| | | | |
|---|---|---|---|
| 0 | 0 | | |
| $\gamma$ | 0 | $\gamma$ | |
| 1 | 0 | $1-\gamma$ | $\gamma$ |
| | 0 | $1-\gamma$ | $\gamma$ |

Since the implicit right-hand side $(\boldsymbol{H}\boldsymbol{q}-\hat{\boldsymbol{q}})/\varepsilon$ has no $\boldsymbol{q}$-component, each implicit stage is solved *pointwise per face*:

$$\hat{\boldsymbol{q}}^{(i)} = \frac{\hat{\boldsymbol{q}}^{*} + k\,\boldsymbol{H}\boldsymbol{q}^{(i)}}{1+k}, \qquad k = \frac{\Delta t\, a^{I}_{ii}}{\varepsilon}. \tag{16}$$

No linear system is solved at all. As $\varepsilon \to 0$ this tends to $\hat{\boldsymbol{q}}^{(i)} = \boldsymbol{H}\boldsymbol{q}^{(i)}$, i.e. the scheme is asymptotic preserving and reduces to an explicit HDG/LMARS (DG) scheme. The time step is limited by the explicit CFL condition $\Delta t \lesssim h/(c(2p+1))$, and the time accuracy is second order.

# 5 Properties

**Proposition 1** (Well-balancing)**.** *Let $p'_h \in P_p$ be element-wise polynomial and globally continuous, $\rho'_h = -\partial_z p'_h/g$ element-wise, $m_h = 0$, and $\hat{\boldsymbol{q}} = \boldsymbol{H}\boldsymbol{q}_h$. Then $\boldsymbol{q}_h$, $\hat{\boldsymbol{q}}$ are a steady state of* (13) *for every $\varepsilon \geq 0$, for both LMARS and LLF (with $\tau_\rho = 0$).*

*Proof.* Because $p'_h$ is continuous, $\hat{p}_f = p'_h(z_f)$ and $\hat{m}_f = 0$, so $\boldsymbol{H}\boldsymbol{q} - \hat{\boldsymbol{q}} = 0$ and the trace equation is steady. In the flux (7), the $\rho'$-row is $\hat{m}\,n + (p'_h - \hat{p})/c = 0$ and the $p'$-row is $c^2\hat{m}\,n + c(p'_h - \hat{p}) = 0$. The $\rho'$-jump enters neither, because the first columns of $\boldsymbol{A}$ and $\boldsymbol{T}$ vanish. Both volume terms of the $\rho'$ and $p'$ equations vanish since $m_h = 0$. The $m$-row of the flux is $\hat{p}\,n + c(m - \hat{m}) = p'_h(z_f)\,n$, and integrating by parts, $(\partial_z v, p'_h)_K - [v\,p'_h]_{\partial K} - g(v, \rho'_h)_K = -(v, \partial_z p'_h + g\rho'_h)_K = 0$. □

**Conservation.** The total mass changes by the sum of the two one-sided mass fluxes on each face. For LMARS this sum is $(2/c_f)(\{\{p'\}\} - \hat{p}) = (2/c_f)\,\varepsilon\,\dot{\hat{p}}$: mass is conserved to round-off for $\varepsilon = 0$, with an $\mathcal{O}(\varepsilon)$ drift otherwise (Table 3). The same holds for $m$ and $p'$. With LLF and $\tau_\rho = 0$ the mass flux is $\hat{m}$, which is single-valued, so mass is conserved exactly for every $\varepsilon$, but $\rho'$ loses one order of accuracy.

**Stability.** For walls and $\varepsilon = 0$ the largest real part of the spectrum of the semi-discrete operator is $\approx 5 \cdot 10^{-15}$ (lapse-rate atmosphere, $N_e = 16$, $p = 2$). With $\varepsilon > 0$ no eigenvalue with positive real part beyond the physical growth of the periodic normal mode was observed.

# 6 Verification

All numbers below come from the Python implementation (`run_tests.py`). The Julia port implements the same algorithm and should reproduce them to 3–4 significant digits.

## 6.1 Convergence

Periodic domain $[0, 2000]\,\mathrm{m}$, isothermal atmosphere ($T_0 = 250\,\mathrm{K}$), exact normal mode $\boldsymbol{q} = \Re(\boldsymbol{v}e^{ikz+\lambda t})$ of (5) on the upward acoustic branch, amplitude $100\,\mathrm{Pa}$, run for one acoustic transit. The SDIRK step is $\Delta t = h/(c(2p+1))$.

LMARS gives the optimal order $p+1$ in all components. The IMEX result for $p = 2$ is limited by the second-order time integrator.

## 6.2 Relaxation error

The error grows linearly in $\varepsilon c/h$ once it exceeds the discretization error. The IMEX floor is the second-order time error.

Table 1: $L^2$ error in $p'$ and observed rates (last refinement), $N_e = 8, 16, 32, 64$.

| flux / method | $p$ | $N_e = 8$ | 16 | 32 | 64 | rate $p'$ | rate $\rho'$ |
|---|---|---|---|---|---|---|---|
| LMARS, DIRK, $\varepsilon = 0$ | 1 | 1.89e+2 | 3.83e+1 | 8.80e+0 | 2.15e+0 | 2.04 | 2.04 |
| | 2 | 8.59e+0 | 1.07e+0 | 1.34e-1 | 1.70e-2 | 3.00 | 3.00 |
| | 3 | 4.09e-1 | 2.59e-2 | 1.62e-3 | 1.01e-4 | 4.00 | 4.00 |
| LMARS, DIRK, $\varepsilon = 10^{-9}h/c$ | 3 | 4.09e-1 | 2.59e-2 | 1.62e-3 | 1.11e-4 | 3.86 | 3.86 |
| LMARS, IMEX, $\varepsilon = 10^{-9}h/c$ | 2 | 9.39e+0 | 1.42e+0 | 2.70e-1 | 6.10e-2 | 2.15 | 2.15 |
| LLF ($\tau_\rho = 0$), DIRK, $\varepsilon = 0$ | 1 | 1.88e+2 | 3.82e+1 | 8.79e+0 | 2.15e+0 | 2.04 | 1.00 |
| | 3 | 4.09e-1 | 2.58e-2 | 1.62e-3 | 1.01e-4 | 4.00 | 3.00 |

Table 2: $L^2$ error in $p'$ versus $\varepsilon c/h$ ($N_e = 16$, $p = 3$); the $\varepsilon = 0$ reference is $2.586 \cdot 10^{-2}$.

| $\varepsilon c/h$ | $10^{-8}$ | $10^{-7}$ | $10^{-6}$ | $10^{-5}$ | $10^{-4}$ | $10^{-3}$ | $10^{-2}$ |
|---|---|---|---|---|---|---|---|
| DIRK | 2.59e-2 | 2.63e-2 | 5.26e-2 | 4.57e-1 | 4.56e+0 | 4.55e+1 | 4.46e+2 |
| IMEX | 4.79e-1 | 4.75e-1 | 4.34e-1 | 4.00e-2 | 4.08e+0 | 4.50e+1 | 4.45e+2 |

### 6.3 Well-balancing

Lapse-rate atmosphere ($T_0 = 288\,\mathrm{K}$, $\Gamma = 6.5\,\mathrm{K/km}$), walls, $[0, 10]\,\mathrm{km}$, $N_e = 40$, $p = 3$. The hydrostatic perturbation is $p' =$ continuous interpolant of $50\cos(2\pi z/7\,\mathrm{km}) + 30(z/10\,\mathrm{km})^2\,\mathrm{Pa}$, with $\rho' = -\partial_z p'/g$. After 200 steps the maximum relative deviation is $3.7 \cdot 10^{-12}$ (DIRK, $\varepsilon = 0$ and $10^{-3}$) and $\leq 10^{-15}$ (IMEX), i.e. round-off.

### 6.4 Acoustic pulse between walls

Same atmosphere, $N_e = 80$, $p = 3$, Gaussian pressure pulse ($100\,\mathrm{Pa}$, width $300\,\mathrm{m}$, centred at $3\,\mathrm{km}$), $t \in [0, 40]\,\mathrm{s}$.

Table 3: Mass drift and trace residual $\max|\boldsymbol{H}\boldsymbol{q} - \hat{\boldsymbol{q}}|$, LMARS.

| case | $\max\lvert\Delta\mathrm{mass}\rvert$ | trace residual |
|---|---|---|
| HDG $\varepsilon = 0$ (DIRK) | 4.0e-14 | 4.3e-12 |
| $\varepsilon = 10^{-4}h/c$ (DIRK) | 1.1e-5 | 4.1e-3 |
| $\varepsilon = 10^{-4}h/c$ (IMEX) | 1.1e-5 | 4.1e-3 |
| $\varepsilon = 10^{-2}h/c$ (DIRK) | 1.1e-3 | 4.0e-1 |

## 7 Code

### 7.1 Files

| | |
|---|---|
| `hdg_relax_euler_gravity.py` | Python solver (NumPy/SciPy) |
| `run_tests.py` | Python tests and plots |
| `julia/HDGRelaxEuler.jl` | Julia module (standard library only) |
| `julia/run_tests.jl` | Julia tests, writes CSV files |
| `julia/plot_results.jl` | optional plots (needs `Plots.jl`) |

### 7.2 Data layout

Element coefficient $(e, r, i)$ (element, variable $\rho', m, p'$, Legendre mode) is stored at $((e-1)\cdot 3 + (r-1))(p+1) + i$ (Julia, 1-based) or $(3e + r)(p+1) + i$ (Python, 0-based). Trace $(f, r)$ is stored

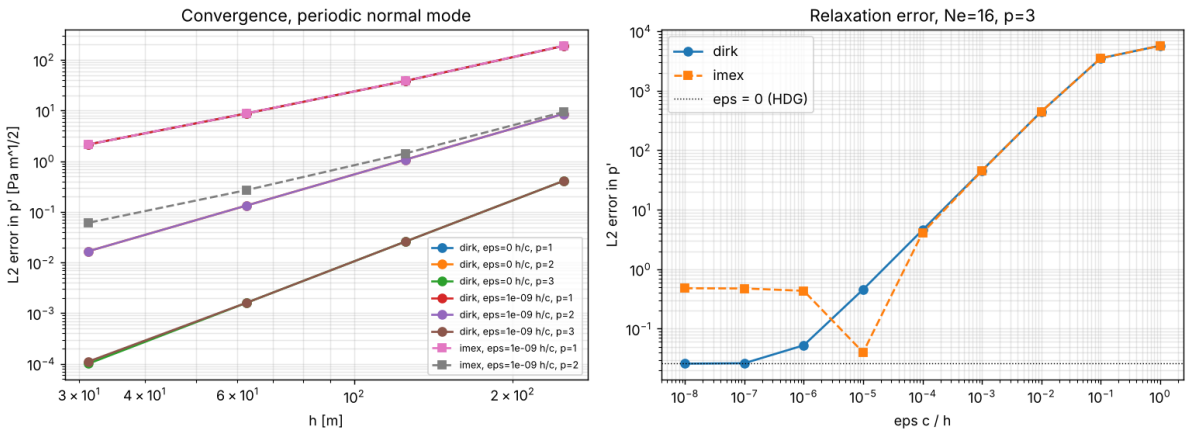



Figure 1: Left: convergence for the normal mode. Right: relaxation error versus $\varepsilon c/h$.

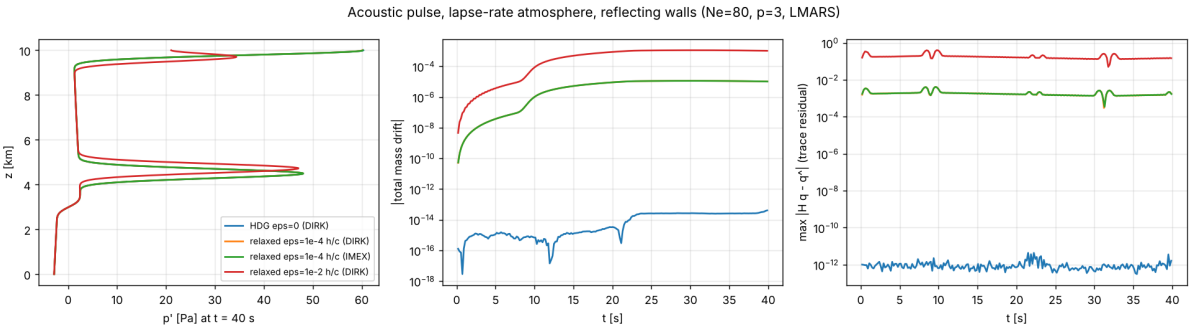



Figure 2: Acoustic pulse with walls: pressure at $t = 40\,\mathrm{s}$, mass drift, trace residual.

at $3(f-1)+r$ (Julia) or $3f+r$ (Python). The state vector is $\boldsymbol{U} = (\boldsymbol{q}, \hat{\boldsymbol{q}})$.

### 7.3 Usage (Julia)

```
include("HDGRelaxEuler.jl"); using .HDGRelaxEuler
atm = Atmosphere(T0 = 288.0, Γ= 0.0065)
s = HDGRelax(80, 3, 0.0, 1.0e4, atm; bc = :wall, flux = :lmars, eps = 0.0)
q0 = project(s, z -> vcat(0 .* z', 0 .* z', (100 .* exp.(-((z .- 3000) ./ 300) .^ 2))'
    ))
U0 = initial_state(s, q0)                    # traces consistent: qhat = H q
U = run!(s, U0, 1e-3, 1000; method = :dirk) # or :imex
z, Q = evaluate(s, U[1:s.nq])                # Q[3,:] = p'
```

The Python API is the same: `HDGRelax(Ne, p, zmin, zmax, atm, bc="wall", eps=0.0, flux="lmars")`, `s.run(U0, dt, n, method="dirk")`.

### 7.4 Main options

| option | values | meaning |
|---|---|---|
| `bc` | wall / periodic | reflecting walls ((9)) or periodic |
| `flux` | lmars / llf | stabilization $\boldsymbol{T}$ |
| `eps` | $\geq 0$ | relaxation time $\varepsilon$ (seconds) |
| `tau_scale` | $> 0$ | multiplies $\boldsymbol{T}$ |
| `tau_rho` | $\geq 0$ | $\rho'$ penalty weight (LLF only) |
| `method` | dirk / imex | monolithic SDIRK4 / local IMEX ARS(2,2,2) |

## References


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